

Glomerular (Nephritic) — Post-Streptococcal GN (PSGN)



1. Poststrep GN

WHAT YOU SEE

POSTSTREP GN banner across fence

WHAT IT ENCODES

Post-streptococcal GN is the prototypical acute nephritic syndrome — it follows infection with nephritogenic strains of group A beta-hemolytic Streptococcus (Streptococcus pyogenes) by a latent interval. It is an IMMUNE-COMPLEX disease: streptococcal antigens deposit in or plant on the glomerulus and trigger complement-mediated injury, producing the nephritic quintet of hematuria, oliguria, edema, hypertension, and mild azotemia. It is the classic 'low complement, normalizes' GN of childhood.

BOARD TRAP

WRONG: diagnosing PSGN during an active sore throat. The discriminator is LATENCY — PSGN follows the strep infection after a 1–3 week (pharyngitis) gap, so hematuria occurring DURING or within 1–2 days of the infection is IgA nephropathy ('synpharyngitic'), not PSGN.

2. M-type nephritogenic strain

WHAT YOU SEE

M-type nephritogenic gangster, SPEB / NAPlr

WHAT IT ENCODES

Only specific NEPHRITOGENIC M-type strains of group A strep cause PSGN, not every strep infection. The key nephritogenic antigens are SPEB (streptococcal pyrogenic exotoxin B / its zymogen) and NAPlr (nephritis-associated plasmin receptor), which localize to the glomerulus, activate complement via the alternative pathway, and recruit immune complexes. This is why most strep infections never cause nephritis — strain-specific virulence factors are required.

BOARD TRAP

WRONG: assuming any streptococcal pharyngitis or any positive throat culture will cause GN. The discriminator is strain nephritogenicity — only specific M-type strains carrying SPEB/NAPlr are nephritogenic, so a recent strep infection alone, without the right strain, does not produce PSGN (and antibiotics for non-nephritogenic strains are irrelevant to nephritis risk).

3. Children most affected

WHAT YOU SEE

Kids on bouncy castle / playground

WHAT IT ENCODES

PSGN classically affects CHILDREN aged 5–12 years and is the most common cause of acute nephritic syndrome in children worldwide. In kids the course is almost always self-limited with excellent recovery (>95% full renal recovery). It is uncommon under age 2. The pediatric predominance and benign pediatric prognosis are core exam points, contrasted with the worse adult/elderly course.

BOARD TRAP

WRONG: extrapolating the excellent pediatric prognosis to adults. The discriminator is age — children recover almost completely (>95%), whereas adults and especially the elderly have substantially higher rates of persistent azotemia, hypertension, and progression to chronic kidney disease/ESKD.

4. Latency 2–3 weeks

WHAT YOU SEE

2 weeks ago strep throat / 6 weeks ago skin infection signs

WHAT IT ENCODES

The hallmark LATENT period reflects the time needed to mount the antibody/immune-complex response: hematuria appears about 1–3 weeks (classically ~2 weeks) after streptococcal PHARYNGITIS, but 3–6 weeks after streptococcal SKIN infection (impetigo/pyoderma). Mnemonic: 'throat is quicker (~2 wk), skin is slower (~4–6 wk).' This delay is the single best historical discriminator from IgA nephropathy.

BOARD TRAP

WRONG: choosing PSGN when hematuria starts within 1–2 days of an upper respiratory infection. The discriminator is the latent interval — gross hematuria SYNCHRONOUS with the pharyngitis (1–2 days) is IgA nephropathy (synpharyngitic), whereas a 1–3 week (throat) or 3–6 week (skin) delay defines PSGN.

5. Hematuria + RBC casts

WHAT YOU SEE

Cola-colored urine cup, hematuria + RBC casts card

WHAT IT ENCODES

The urinary hallmark is COLA/TEA-colored (smoky brown) urine from glomerular bleeding, with DYSMORPHIC RBCs and RBC CASTS on microscopy — the defining nephritic sediment. There is usually sub-nephrotic proteinuria. RBC casts confirm the bleeding is glomerular (formed within the tubular lumen) rather than from the lower urinary tract.

BOARD TRAP

WRONG: attributing cola-colored urine with RBC casts to a urologic source or to gross hematuria of bladder origin. The discriminator is the sediment — dysmorphic RBCs and RBC casts localize bleeding to the glomerulus, ruling out lower-tract causes (stone, tumor, cystitis) that give isomorphic RBCs without casts.

6. Periorbital puffiness

WHAT YOU SEE

Puffy facial 'bital' puffiness, edematous boy

WHAT IT ENCODES

Periorbital edema (puffy eyes, especially in the morning) and generalized edema arise from renal SALT and WATER retention — the inflamed glomeruli drop GFR, the kidney avidly retains sodium, and the resulting volume overload expands the extracellular space. This is a nephritic, volume-driven edema (low urine output, hypertension) rather than the hypoalbuminemic oncotic edema of nephrotic syndrome.

BOARD TRAP

WRONG: interpreting the edema as nephrotic (hypoalbuminemic) edema. The discriminator is mechanism — PSGN edema is from PRIMARY salt/water retention and volume overload (so it comes with hypertension and only sub-nephrotic proteinuria), not from massive proteinuria and low albumin as in nephrotic syndrome.

7. Hypertension

WHAT YOU SEE

School nurse BP cuff 'not normal'

WHAT IT ENCODES

Hypertension is a core nephritic feature driven by VOLUME overload from sodium/water retention (not primarily renin-mediated). It can be severe enough to cause hypertensive encephalopathy or seizures in children. Because the mechanism is volume, management centers on salt/water restriction and LOOP DIURETICS (eg furosemide), with vasodilators (eg calcium-channel blockers) added if needed.

BOARD TRAP

WRONG: reaching first for an ACE inhibitor in acute PSGN hypertension. The discriminator is that the HTN and edema are VOLUME-mediated, so loop diuretics and salt restriction are first-line; ACEi/ARBs are relatively avoided acutely given the risk of hyperkalemia and worsening GFR in the setting of acute kidney injury.

8. Subepithelial humps (EM)

WHAT YOU SEE

A label pointing to dome-shaped 'humps' sitting on top of the membrane — the bumps perched on the outer GBM surface are the pathognomonic subepithelial immune-complex humps seen on EM.

WHAT IT ENCODES

Electron microscopy shows pathognomonic SUBEPITHELIAL 'humps' — large dome-shaped electron-dense immune deposits sitting on the outer (epithelial/podocyte) side of the GBM. Immunofluorescence shows GRANULAR (lumpy-bumpy) deposits of IgG and C3 in a 'starry sky' pattern, and light microscopy shows a diffuse hypercellular, endocapillary proliferative glomerulus. The triad (LM endocapillary proliferation, IF granular starry-sky, EM subepithelial humps) is the classic PSGN signature.

BOARD TRAP

WRONG: confusing granular subepithelial humps with mesangial IgA deposits. The discriminator is the IF pattern and deposit site — PSGN shows GRANULAR ('lumpy-bumpy') IgG/C3 with subepithelial humps on EM, whereas IgA nephropathy shows MESANGIAL IgA deposits; the IF/EM picture cleanly separates the two delayed-vs-synchronous hematuria diseases.

9. Hypercellular glomerulus

WHAT YOU SEE

HYPERCELLULAR bouncy castle glomerulus

WHAT IT ENCODES

Light microscopy shows DIFFUSE ENDOCAPILLARY PROLIFERATIVE glomerulonephritis — enlarged, hypercellular glomeruli with proliferating endothelial and mesangial cells PLUS infiltrating neutrophils filling the capillary loops (an exudative, 'busy' glomerulus). This endocapillary, neutrophil-rich hypercellularity is the LM correlate of acute PSGN and distinguishes it from purely mesangial diseases.

BOARD TRAP

WRONG: calling this mesangioproliferative GN. The discriminator is the COMPARTMENT of hypercellularity — PSGN floods the capillary LOOPS with neutrophils and endothelial cells (endocapillary/exudative), whereas mesangioproliferative GN/IgA crowds only the central MESANGIAL stalk with normal complement.

10. Low C3 complement

WHAT YOU SEE

A lab sign showing complement/C3 dropped LOW — the low reading recovers by 6-8 weeks, so persistently low C3 means it is NOT simple PSGN.

WHAT IT ENCODES

Serum C3 is LOW during the acute phase (complement consumed largely via the ALTERNATIVE pathway), typically with relatively preserved C4. The critical exam point is the TIME COURSE: C3 normalizes within about 6–8 weeks. Persistently low C3 beyond 8–12 weeks is a red flag that should redirect you to MPGN/C3 glomerulopathy or lupus, prompting biopsy.

BOARD TRAP

WRONG: expecting normal complement, OR ignoring the recovery timeline. The discriminator is that C3 is LOW acutely and RECOVERS by 6–8 weeks — normal complement argues against PSGN, while C3 that stays low beyond 8 weeks means it is NOT simple PSGN (think C3 glomerulopathy/MPGN or lupus nephritis).

11. Streptozyme / ASO positive

WHAT YOU SEE

Streptozyme 80–95%+ trophies

WHAT IT ENCODES

Evidence of recent strep infection confirms the diagnosis: anti-streptococcal antibodies — ASO (antistreptolysin O), anti-DNase B, anti-hyaluronidase — and the combined streptozyme panel, which is positive in ~80–95% of cases. ASO rises after PHARYNGITIS, whereas anti-DNase B is the more reliable marker after SKIN infection (impetigo). A throat culture is often negative by the time nephritis appears because of the latent period.

BOARD TRAP

WRONG: relying on ASO alone after a skin infection, or expecting a positive throat culture at presentation. The discriminator is the right serology for the right site — use anti-DNase B (not ASO) to confirm post-impetigo PSGN, and remember the throat culture is frequently negative weeks later, so serology, not culture, establishes recent strep.

12. Treatment does not prevent

WHAT YOU SEE

A trash can with a discarded 'no role' / 'too late' sign reading DOES NOT PREVENT NEPHRITIS — antibiotics are tossed aside because, unlike rheumatic fever, treating the strep does not stop the kidney disease already underway.

WHAT IT ENCODES

Antibiotic treatment of the strep infection does NOT prevent PSGN (unlike acute rheumatic fever, which antibiotics DO prevent) — by the time nephritis develops the immune-complex process is already underway. Management is therefore SUPPORTIVE: sodium/water restriction, loop diuretics for edema/hypertension, and BP control; antibiotics are given to eradicate residual strep and limit spread but not to alter the renal course. Most children recover fully.

BOARD TRAP

WRONG: claiming antibiotics prevent PSGN the way they prevent rheumatic fever. The discriminator is the disease — prompt penicillin prevents acute RHEUMATIC FEVER but does NOT prevent post-streptococcal GN, so PSGN treatment is supportive (diuretics, salt restriction, BP control), not antibiotic-driven nephroprotection.

13. Elderly worse prognosis

WHAT YOU SEE

Elderly azotemia 60% ESKD, stretcher

WHAT IT ENCODES

Prognosis is age-dependent: CHILDREN recover almost completely (>95% full recovery, supportive care suffices), but ADULTS and especially the ELDERLY do worse — higher rates of persistent azotemia, hypertension, residual proteinuria, and progression to chronic kidney disease/ESKD (the figure on the scene cites ~60% azotemia/ESKD risk in the elderly). Age at onset is the dominant prognostic factor.

BOARD TRAP

WRONG: reassuring an elderly diabetic patient that PSGN will fully resolve like it does in children. The discriminator is age (and comorbidity) — pediatric PSGN is benign and self-limited, whereas elderly/adult patients carry a markedly higher risk of incomplete recovery and progression to CKD/ESKD, warranting closer follow-up and lower threshold for biopsy if atypical.
